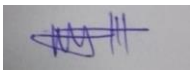
MAKERERE UNIVERSITY

PROJECT WAY-GUARD

TECHNICAL REPORT

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DECLARATION

We, Team Avron, declare that this report is the product of our own work for the course Biomedical Engineering Design I. The needs, observations, screening decisions and analyses described here were produced by our team during fieldwork and team sessions. Wherever we relied on ideas, data or statements from other authors, we have acknowledged them in the text and listed them in the References section.

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ACKNOWLEDGEMENT

We thank God for the health, strength and focus that carried us through this semester.

We are grateful to our tutors, Dr. Robert Ssekitoleko and Ms. Nalwoga Racheal, , whose feedback repeatedly sharpened our thinking about what makes a clinical need worth solving.

We also thank the professionals who gave us their time and honest perspectives: Laymond Kalema (laboratory technologist), Nakalanzi Sauds (midwife, Kawempe National Referral Hospital) and Eng. Allan Rodin Gidoi (biomedical engineer, Ruby Hospital). Their accounts of daily practice grounded our work in real clinical conditions.

ABSTRACT

Many promising medical technologies fail because they begin with a solution instead of a well-understood need. This report documents how Team Avron applied the Identify phase of the Stanford Biodesign process to find and prioritise clinical needs that a small student team could address within one academic year in Uganda. We selected four strategic focus areas (maternal and neonatal child health, laboratory medicine, surgery and rehabilitation), reviewed background literature, prepared interview guides, and gathered observations and interviews from a laboratory technologist, a midwife and a biomedical engineer. The findings were written up as need statements, each naming a population, a problem and an outcome. These statements passed through four screening stages: a regulatory and safety filter, a prototyping and technical feasibility filter, the Stanford 2x2 importance-versus-satisfaction matrix, and a weighted multi-attribute decision matrix. The process produced a final top ten and, from it, three prioritised needs: real-time hazard detection for visually impaired people navigating outdoor environments, rapid blood glucose screening for patients at risk of hypoglycaemia, and affordable simulation tools for cardiac surgical trainees. Concept generation was not reached during this semester and is set out as the next stage of work.

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LIST OF ABBREVIATIONS AND ACRONYMS

Abbreviation	Meaning
BME	Biomedical Engineering
CPAP	Continuous Positive Airway Pressure
CVD	Cardiovascular disease
FDA	United States Food and Drug Administration
GBD	Global Burden of Disease
IMU	Inertial measurement unit
LMIC	Low- and middle-income countries
MAUT	Multi-attribute utility theory
NICU	Neonatal intensive care unit
POC	Point of care
WHO	World Health Organization

CHAPTER ONE: INTRODUCTION

1.1 Background

Biomedical engineering design is most effective when it starts from the clinic rather than the workshop. The Stanford Biodesign process formalises this idea: innovators first immerse themselves in clinical settings, record what they observe, and only then commit to a problem worth solving (Yock et al., 2015). A good need is specific, can be observed, and is linked to a result we can measure. It is chosen before anyone thinks of a solution.

In Uganda and similar places, clinicians often work with equipment that is scarce, costly to maintain, or made for very different conditions. This affects what a realistic device can be. A concept that needs imported consumables, steady electricity or a specialist operator may look clever but be useless in a district hospital. For this reason, Team Avron treated local feasibility, cost and safety as part of the problem from the first day of screening.

Visual impairment is a large problem. The Global Burden of Disease study estimated that about 43.3 million people were blind worldwide in 2020, and most of them live in low- and middle-income countries (GBD 2019 Blindness and Vision Impairment Collaborators, 2021; World Health Organization, 2019). Assistive products are meant to close the gap that a disability leaves, but they do not reach most of the people who need them. WHO and UNICEF (2022) report that more than 2.5 billion people need at least one assistive product, nearly one billion of them are denied access, and access can be as low as 3% of the need in the poorest countries.

Uganda shows the same pattern. A rapid assessment of avoidable blindness (RAAB) in four districts of Western Uganda found that, among adults aged 50 years and above, 0.9% were blind, 0.9% had severe vision impairment and 4.5% had moderate vision impairment. Untreated cataract caused 49.4% of the blindness (Bondok et al., 2025). The survey covered one region and older adults only, so it cannot describe the whole country. Still, it shows that a noticeable share of older adults **lives** with serious sight loss, and that the leading cause of blindness, untreated cataract, is one that can be avoided.

For people who stay blind or visually impaired, getting around outside is one of the hardest daily tasks. The white cane is the usual aid in developing countries (Chanana, 2020), but it warns the user only about low obstacles that are close by (Singh et al., 2010). Devices that sense more do exist, but they are expensive and are often designed for conditions very different from ours (Kim & Cho, 2013; Chanana, 2020). Our first need starts from this gap.

Why needs come before solutions. A recurring lesson in medical device development is that the most common reason a technology fails is not poor engineering but a poor choice of problem. Teams fall in love with a clever mechanism and then search for a clinic that might use it. By the time they discover that the clinical problem is minor, already well served, or impossible to fit into the daily routine of a ward or laboratory, months of effort have been spent. The Biodesign approach reverses this order. It asks the team to stay in the clinic long enough to see what clinicians actually struggle with, to write that struggle down as a neutral statement that contains no solution, and to test the statement against evidence before any drawing or prototype begins. For a student team

with one academic year, this discipline is not a luxury. It is the main protection against spending the year on the wrong thing.

What makes a need worth pursuing. Not every difficulty observed in a clinic is a good need. A worthwhile need is tied to a defined group of patients or clinicians, describes a problem that happens often enough or harms seriously enough to matter, and links to an outcome that can be recognised and measured, such as fewer injuries, faster results or more trainees reaching competence. It should also be a problem that current practice solves poorly. If a satisfactory solution is already affordable and available, then another device adds little. Finally, a need should be addressable by a small team with realistic resources. A problem that requires a national policy change, a new drug or a billion-dollar factory may be important, but it is not a design project for students. Each of these ideas shaped the questions we asked during fieldwork and the criteria we used later when we screened our list.

The Ugandan health system context. Uganda has a tiered public health system that runs from village health teams and health centres at the base, through district and regional referral hospitals, to national referral hospitals at the top. Alongside the public system there are private not-for-profit facilities, often run by faith-based organisations, and private for-profit hospitals and clinics. Patients and clinicians move between these levels, and equipment is distributed unevenly among them. A national referral hospital may have specialised machines that a district facility has never seen, and a health centre may rely on a handful of basic instruments and the skill of a few staff. Any need we chose therefore had to be considered against this range of settings, because a device that works well in a well-equipped referral hospital may still fail to reach the places where the need is greatest.

Equipment, maintenance and the supply chain. Many clinicians and biomedical engineers describe the same pattern: equipment arrives through donations or large procurement programmes, is used for some time, and then stops working because a part, a consumable or a service visit is not available. Devices designed for temperate, clean, air-conditioned environments can struggle with dust, heat, humidity and unstable electricity. Where imported spare parts take months to arrive, a small fault can take a machine out of service for a long time. This is why we treated maintainability as a design consideration from the start. A device that local technicians can understand, repair and recalibrate with locally available parts has a far better chance of staying in use than one that depends on a foreign manufacturer. It is also why we paid attention to consumables, since a device that needs an expensive single-use item for every patient can be unaffordable even when the main unit is cheap.

Electricity, connectivity and the operator. Three further assumptions are often built into imported devices without being stated. The first is a steady power supply, which cannot be taken for granted in many facilities and homes. Power cuts and voltage swings are common, and a device may need a battery or a low-power design to stay useful. The second is reliable connectivity. Features that depend on constant internet access or cloud services may not work in rural settings or may be too costly for users to keep running. The third is the availability of a trained operator. A technology that needs a specialist to set up, calibrate or interpret results may be unusable where

such a specialist does not exist. Designing for the reality of the setting means making each of these assumptions explicit and choosing, wherever possible, an approach that does not depend on them.

Cost, affordability and who pays. Affordability has more than one meaning. There is the purchase price of a device, the running cost of consumables and batteries, the cost of repair, and the cost of training people to use it. There is also the question of who pays. In public facilities, the budget is limited and fixed, and equipment is often bought centrally or arrives through donors. In the private sector, a device must earn back its cost through fees. For assistive products, the user or the family often pays out of pocket, and for many households even a modest price is a serious barrier. A design that looks inexpensive in a laboratory can still be out of reach once all of these costs are added. For this reason, our screening criteria later included cost and local feasibility explicitly, and not only clinical importance.

Safety, regulation and ethics. Medical technologies, even simple ones, can cause harm if they fail or are misused. A glucose reading that is wrongly high could hide a dangerous low, and a hazard alert that does not sound could lead a user into danger. Regulators and standards bodies exist to set expectations for safety and performance, and a device intended for real patients eventually has to meet them. At the needs-finding stage, regulatory thinking is not about paperwork; it is about asking early whether a need could be addressed without creating a risk that a student team cannot manage. We also considered ethics in our fieldwork. We asked clinicians for their time and honest opinions, we took care not to disrupt patient care, and we did not collect personal patient information. These principles guided how we behaved in each facility and how we wrote up what we learned.

The burden of vision loss beyond the numbers. The statistics on blindness and vision impairment describe how many people are affected, but they say less about how a person lives with the condition. Losing sight, or never having it, changes how a person learns, works, travels and takes part in community life. Children who are visually impaired may find it harder to attend school and move around campus. Adults may lose paid work or find it harder to find it, and older people may become dependent on relatives for tasks they once did alone. The effects are not only physical. People who fear falling or colliding with obstacles may choose to stay at home, which reduces social contact and can affect mental health. Seen this way, a tool that helps a person move safely outdoors is not only a safety device. It can support education, employment and dignity, which are the outcomes that matter most to the people concerned.

Why outdoor movement is especially demanding. Outdoor environments are more complex than indoor ones. Surfaces change from smooth to broken, kerbs and drains appear without warning, vehicles share the road with pedestrians, and vendors, parked motorcycles and low-hanging branches occupy spaces that a walker would expect to be clear. In many towns and cities, the footpath is not a continuous, protected space, and people on foot must make decisions about when and where to cross or step aside. A person who is visually impaired has to gather information through touch, hearing and memory, and has to do so continuously. The cane helps with what is directly ahead and near the ground, but it does not give advance warning of objects above the waist or of fast-moving hazards. This helps explain why the first need we chose focuses on detecting several kinds of hazards in time for the user to respond.

Health needs that do not announce themselves. Some clinical problems are obvious, such as a broken bone, while others develop silently and are detected only when someone measures them. Low blood glucose belongs to the second group. A patient, especially a young child or a newborn, may not be able to describe symptoms, and the signs can resemble other conditions. Without a quick and dependable test, the problem may be missed until harm has occurred. Problems of this type depend heavily on the availability of screening tools at the point where care is given, and not days later in a distant laboratory. This is one reason we looked closely at how rapid, repeatable testing could be delivered in settings where laboratory services and supplies are limited.

Training the next generation of clinicians. Not every need is about a patient directly. Some are about the people who care for patients. Surgery is a craft learned through repetition, supervision and gradually increasing responsibility. Where few experienced surgeons are available and training positions are scarce, there are fewer chances to practise before operating on real patients. Simulation offers a safer way to build basic skills, but good simulators are costly and many training programmes cannot buy them. A need that concerns the education of surgeons may seem less urgent than one concerning a patient in an emergency, yet the effect is large, because a single well-trained surgeon can treat many patients over a career. This reasoning helped us to see why a training-related need could sit alongside clinical needs on our final list.

The role of a student team. A student design team has real strengths and real limits. Its strengths are curiosity, time to read widely, the freedom to ask basic questions without commercial pressure, and access to university laboratories, workshops and tutors. Its limits are a small budget, limited access to patients, little experience with regulation and only one academic year to complete the work. Because of these limits, we aimed at needs where a modest prototype could demonstrate a meaningful improvement, and where the main risks could be understood and tested without clinical trials. We also accepted that our work would be a starting point. A good needs statement, with the evidence to support it, can be used by later teams, partners or clinicians who may take the work further.

Teamwork and division of labour. Our team organised its work around defined roles: a team lead who kept the schedule and coordinated communication, research leads who guided the reading and the interview guides, and a secretary who kept records of meetings and decisions. Roles did not prevent everyone from contributing to every stage. Fieldwork, clustering of observations, drafting of need statements and screening were done in joint sessions so that decisions reflected a shared view. Keeping written records mattered a great deal. When we later had to justify why a need was removed at a particular filter, we could go back to the notes and the criteria instead of relying on memory.

How the rest of this report is organised. Chapter One gives the background, the literature on the three needs that emerged from our screening, the problem statement, and the objectives and scope of the report. Chapter Two describes the Stanford Biodesign process, our needs-finding activities, and the four filters through which we reduced our list to the final top ten and then to three priority needs. Chapter Three discusses what we learned, sets out the plan for concept generation, and concludes the report. The references follow, and the appendix contains the master list of refined need statements for readers who wish to see the full range of needs that were considered. Readers

who are interested mainly in the outcome may turn directly to Section 2.3.7, while those who wish to follow the reasoning from the start should read the chapters in order.

What this background does and does not claim. It is important to be clear about the limits of the material above. The general observations about health systems, equipment and affordability describe patterns that clinicians, engineers and researchers commonly report, and they were reinforced by what we heard in our own interviews. They are not new research findings, and they do not apply in the same way to every facility. The figures quoted from published sources come from specific studies, regions and populations, and we have taken care to describe them as such. Where we have only our own reasoning, for example about likely stakeholders, we have said so in the relevant sections. This honesty about evidence is part of the approach we followed, because a need that rests on weak evidence is one that is likely to be dropped when it is screened.

Reading the local picture through three lenses. To keep our thinking organised, we looked at every possible need through three lenses: the patient or user, the clinician or caregiver, and the health system. The patient or user lens asks who experiences the problem and what it costs them in health, safety, time and money. The clinician or caregiver lens asks how the problem shows up in daily work, which workarounds are used, and what would make the job safer or faster. The health system lens asks whether a solution could be bought, maintained and supported by the facility or programme that would use it. A need that looked strong through one lens but weak through another was usually flagged for closer discussion. For example, a device could be useful to clinicians but impossible for the facility to maintain, and in such a case we had to decide whether the need was truly addressable by a student team.

Lessons from the field before we started. Even before our first interview, our reading and conversations with tutors taught us several lessons that we carried into fieldwork. First, clinicians are busy, and an interview that respects their time is more likely to produce honest and detailed answers. Second, the best observations often come from asking about a recent specific case instead of asking for general opinions. Third, people tend to describe solutions they have heard of, so questions must be steered back to the underlying problem. Fourth, a workaround is a signal. When a clinician has built a habit or a makeshift tool to get around a limitation, the limitation is usually worth recording as a possible need. These lessons helped us to listen for problems and not for suggestions.

Why we paid attention to existing solutions early. A need can only be judged properly if we know what already exists. Part of the purpose of the literature review was to identify existing products, published prototypes and common practice for each of the three needs, so that we could see where the gap really lies. In the case of mobility aids for visually impaired people, a great deal of technology exists, from ultrasonic canes to wearable sensors, yet the problem of affordability, weight and reliability in unstructured outdoor settings remains open. In the case of glucose screening, accurate tests exist, but supply and cost can be barriers in low-resource facilities. In the case of surgical training, simulators exist, but they are expensive. In each case, the gap is not the absence of an idea but the absence of a solution that fits the setting, and this became a theme that ran through our screening.

Measurable outcomes and how we thought about them. Biodesign asks teams to attach an outcome to every need, and we found this a useful habit. An outcome gives a direction for later testing, because it tells us what improvement a successful device should produce. For hazard detection, the outcome is independent travel without injury, which could be assessed by counting collisions, near misses or the distance a user covers confidently. For glucose screening, the outcome is the prevention of neurological damage, which depends on how quickly a low value is found and treated. For surgical training, the outcome is the competence of trainees in fundamental skills, which can be assessed by observed performance on standard tasks. Writing these outcomes at this early stage does not commit us to a particular solution, but it keeps us honest about what success would mean.

Looking ahead. The background above is meant to prepare the reader for what follows. It explains the way of thinking that guided our work, the conditions in which our needs would have to function, and the reasons we gave weight to affordability, local feasibility and safety. It also explains why our work is presented as the first half of a longer journey. The needs identified here are not finished products or finished designs. They are carefully selected starting points, supported by literature and fieldwork, that a later stage of work can turn into concepts, prototypes and, eventually, devices that serve patients and clinicians in settings like our own.

A closer look at mobility and independence. Independent mobility is more than the ability to move from one place to another. It includes the confidence to leave home without waiting for a companion, the ability to choose a route, and the freedom to change plans. For many visually impaired people, planning a journey means learning landmarks, counting steps, listening for traffic and relying on memory of the route. A sudden change, such as a parked vehicle, roadworks or a fallen branch, can break this mental map and create danger. A device that provides early and clear warnings can reduce the mental effort of travel and allow the user to concentrate on direction and surroundings. For this reason, the quality of the warning matters as much as the sensing itself. An alert that is late, confusing or too frequent may be ignored, and an alert that cannot be understood in a noisy street is of little use.

Why a simple signal is often better than a complex one. In noisy outdoor settings, users depend on their hearing to detect traffic, voices and other cues about their surroundings. Any device that covers the ears or fills the sound space with beeps can interfere with this natural awareness. Touch-based feedback, such as vibration, does not compete with environmental sound, but it too has limits, because it can be hard to feel through clothing or over long periods. A good design therefore has to balance the amount of information given with the effort needed to understand it. This is a design question for the concept stage, but we mention it here because it explains why the literature on assistive devices pays so much attention to feedback methods and user comfort, and not only to sensors.

Newborn care and the importance of timing. In the first days of life, a newborn must adjust from receiving a steady supply of nutrients through the placenta to feeding by mouth. During this period, blood glucose can fall, especially in babies who are born early, are small for their age, are unwell, or are not feeding well. Early feeding and careful monitoring are standard ways of preventing and detecting low glucose. Where staff numbers are small and wards are crowded,

however, regular checks can be difficult to carry out. A test that is quick, needs only a small sample and gives a clear result at the bedside can make monitoring more practical. Such features were therefore important when we considered how a screening tool could serve midwives and nurses in busy maternity and neonatal units.

The place of the laboratory. Laboratory services support almost every part of modern medicine, from diagnosing infection to guiding treatment of chronic disease. In many facilities, however, laboratory capacity is limited by equipment, reagents, power and trained staff. When a test cannot be done on site, samples must be sent elsewhere and results may arrive late, or not at all. Point-of-care testing aims to reduce this delay by bringing simple, reliable tests to the patient. Such tests have to be designed carefully, because they are used by staff with different levels of training and in conditions that are less controlled than a central laboratory. Our conversation with a laboratory technologist helped us to appreciate how much depends on supplies, calibration and routine quality checks.

Surgery, skills and safe practice. Surgical skill is built through a mix of knowledge, repeated practice and feedback. In well-resourced programmes, trainees can practise on simulators, cadavers and animal or synthetic models before operating on patients, and they can receive structured feedback on their technique. In programmes with fewer resources, trainees often learn mainly by watching and assisting, with fewer opportunities to practise basic tasks such as suturing, handling delicate tissue and working in a small operating field. Simulation does not replace experience, but it can make early practice safer and more frequent. Affordable and reproducible models that can be built and maintained locally could therefore widen access to this kind of practice, which is the idea behind the third need on our final list.

Rehabilitation and the wider picture of function. Rehabilitation aims to help people regain or maintain function after illness, injury or disability. It involves therapists, doctors, nurses, engineers and often the family. Assistive products, such as mobility aids, prostheses and supports, are a major part of this work. Yet, as the figures from WHO and UNICEF show, a very large share of people who need such products do not receive them. The reasons include cost, lack of trained providers, poor supply and distance. This wider picture helped us to understand why rehabilitation was included among our strategic focus areas, and why the first need on our final list, which concerns an assistive solution, sits comfortably within that area.

Why we valued evidence from more than one source. Throughout the project, we tried to support each important point with more than one kind of evidence: published studies, reports from international organisations, and the accounts of practising clinicians. Each source has strengths and limits. Published studies provide measured data but may describe a different setting. Reports from international organisations give a broad view but may not capture local detail. Clinicians describe the daily reality of care but speak from their own experience. By placing these side by side, we hoped to avoid relying too heavily on any one of them. Where they agreed, we gained confidence in a need, and where they differed, we noted the difference and treated it as something to be explored further.

Assumptions we tried to make visible. Every design project begins with assumptions, and the risk lies in not noticing them. In our case, we assumed that the problems described in the literature also occur in Uganda, that clinicians are willing to adopt tools that make their work easier, and that users would value a device that is affordable and simple more than one that is feature-rich. Some of these assumptions are supported by what we heard in interviews, while others still need to be tested with the people concerned, particularly the visually impaired users who are the focus of our first need. Writing these assumptions down is useful because it shows what must be checked first when the project moves on.

The value of a documented process. A final reason for explaining our approach in some detail is that a clear process can be repeated, criticised and improved. Another team could take our focus areas, follow the same steps and arrive at a different list of needs, and that would not be a failure. It would show that the process is open to different conditions and different judgements. What matters is that each decision is written down along with the reason for it, so that readers can follow the path from observation to priority. We hope that the background presented here, together with the chapters that follow, gives the reader enough information to judge our choices for themselves.

Summary of the background. In summary, the background to our work is a health system in which clinicians are capable and committed but often limited by equipment, supplies, power and cost. In that setting, good design starts with a well-understood need, is shaped by affordability, maintainability and safety, and is tested against evidence from literature and from people who work in the field. The three needs that came out of our screening, hazard detection for visually impaired people, rapid glucose screening, and affordable surgical simulation, each reflect a gap between what technology can do and what is available in practice in our setting. The sections that follow show how these needs were found, screened and selected.

Stakeholders in the wider health setting. Every clinical need involves more people than the patient and the clinician. Administrators decide what equipment is purchased, biomedical engineers and technicians keep it running, procurement officers handle supply, and trainers and educators shape how new tools are taught. Families and communities provide care and support, often at their own expense. When we considered each need, we tried to ask who would use the solution, who would maintain it, who would pay for it, and who might be affected by it in ways we had not expected. This habit of looking across the whole system helped us to avoid needs that looked attractive from one point of view but would have been blocked by another.

Time, workload and the realities of a busy clinic. Clinical work is shaped by time. A midwife attending to several mothers, a laboratory technologist processing a queue of samples, and a surgeon moving between theatre and ward all work under pressure. Any device that adds steps, takes long to set up or needs extra paperwork is likely to be set aside. During our interviews, we listened for these constraints, because a technically correct solution that does not fit the workflow will not be used. This is why we looked for needs where a device could save time or reduce effort, and not only improve accuracy.

Training and local capacity to use new tools. The success of any new tool depends partly on the people who use it. Training takes time and money, and staff move between facilities. Tools that

are intuitive, with clear indications and few steps, are easier to teach and less likely to be misused. For devices intended for the public, such as assistive aids, the same principle applies: the user must be able to learn it quickly and trust it. We therefore treated simplicity not as a lack of ambition but as an important design goal, particularly in settings where long training courses are not realistic.

Sustainability beyond a single project. Many well-meant technology projects end when the funding or the team moves on. To reduce this risk, we considered from the beginning whether a solution could be made from locally available components, repaired by local technicians, and documented well enough for others to continue. Sustainability also includes environmental concerns such as the disposal of batteries and single-use items. These concerns are easier to address in the early stages of design than after a product is in use, and they are part of the reason we asked about consumables and maintenance during our fieldwork.

Learning from what has been tried before. The history of medical technology in low-resource settings includes both successes and disappointments. Successful examples tend to share features: they respond to a clearly felt need, they are priced within reach, they work with the local infrastructure, and they are supported by training and spare parts. Disappointing examples often involve devices that were donated without support, were too complex to maintain, or were designed without input from the people who would use them. Taking these lessons seriously shaped how we selected and screened needs, and how we plan to involve users and clinicians in later stages.

Communication with clinicians and participants. Our contacts with clinicians were built on courtesy and clarity. We explained the purpose of the course, made it clear that the interviews were for learning, and asked permission before recording notes or taking photographs. We thanked each participant for their time and shared with them the aim of the project. Good communication in the field matters for ethical reasons and also for practical ones: clinicians who feel respected are more willing to share detailed accounts of their work, including the frustrations and workarounds that point to real needs.

Documentation and traceability of decisions. From the start, we agreed that every need statement, every screening decision and every score would be recorded with its reasons. This traceability allowed us to review decisions as a team, to correct mistakes when we found them, and to explain our choices to our tutors. It also means that the master list of need statements in the appendix, and the tables in Chapter Two, can be read as a record of how our thinking developed, from a broad list of possible problems to a short list of priorities.

Why the three needs differ so much. The three priority needs come from different fields, namely assistive technology, point-of-care diagnostics and surgical education. This variety is a result of our method and not a lack of focus. We began with four strategic areas and allowed the screening criteria to decide which needs rose to the top. As a result, the final needs reflect those where the evidence, the feasibility and the possible impact were strongest in our assessment. At the same time, they share common features: each addresses a gap between what exists and what is affordable or usable in our setting, and each is specific enough to be tested.

A note on language and terms. We have tried to use plain language where possible and to define technical terms when they first appear. The list of abbreviations at the start of the report is provided for easy reference. Terms such as need statement, screening, filter and decision matrix are used in the specific senses explained in Chapter Two. Readers who are new to the Biodesign process may find it helpful to read Section 2.1 before the rest of the chapter, because it explains how the Identify, Invent and Implement phases relate to one another and where our work fits among them.

This report covers the first half of our design journey. It explains how we found needs, how we cut a large list down to a few, and why we chose the needs we did.

1.2 Literature review

The literature review below covers the three needs that emerged from our screening. It is not exhaustive. Its purpose is to show that each need is real, significant and incompletely served by existing solutions.

1.2.1 Mobility and hazard detection for people who are visually impaired

The Global Burden of Disease study estimated that about 43.3 million people were blind worldwide in 2020, and most of them live in low- and middle-income countries (GBD 2019 Blindness and Vision Impairment Collaborators, 2021; World Health Organization, 2019). Singh et al. (2010) put it more strongly: almost 90% of blind people live in developing countries, and most of them live below the poverty line. For these people, moving safely from one place to another is one of the hardest parts of daily life. Messaoudi et al. (2022) describe how visually impaired people struggle to go outdoors and to protect themselves from moving and stationary objects, and how this lowers their confidence because their movement is limited.

The white cane is still the main mobility aid because it is cheap and dependable. Chanana (2020) reports that it is the most widely used aid in developing countries. But it has clear limits. Singh et al. (2010) explain that it detects obstacles only up to knee level and only within about 1 m of the user, so raised obstacles and fast vehicles are noticed too late. Manduchi and Kurniawan (2011) found that about 13% of blind and legally blind adults collide with head-level obstacles once a month or more, and that about 23% of such accidents lead to medical consequences. Kim and Cho (2013) add that cane users find it hard to detect protruding bars and moving vehicles until they are dangerously close, which can lead to collisions and falls.

The hazards that matter outdoors are of different kinds. Sin et al. (2024) give tree branches and tables as examples of head-level and trunk-level obstacles that a cane misses. Nazri et al. (2019) built water detection into their device because puddles are another outdoor hazard for visually impaired walkers. Hertel et al. (2019) say that navigation and obstacle avoidance, especially for head-level objects, are major concerns for visually impaired people. This suggests that the user needs to be warned about hazards at ground, body and head height, and not only about what is directly in front of the feet.

Where the person walks also matters. Chanana (2020) points out that most assistive devices come from developed countries and are designed for structured environments, so they often fail in the unstructured environments found in countries like India. Singh et al. (2010) add that devices sold

internationally are unaffordable for most users, since most blind people live in developing countries. We expect the same to be true on Ugandan roads and footpaths, but we have not yet confirmed this with users (see Section 3.1).

Researchers have tried several ways of solving this problem. Smart canes add sensors to the cane itself. The STIC cane uses an ultrasonic sensor for head-level obstacles and a laser sensor for ground-level ones, and warns the user through vibration (Hertel et al., 2019). A retractable ultrasonic add-on detects head-level and trunk-level obstacles within 200 cm and 300 cm (Sin et al., 2024). The Smart e-Stick adds ultrasonic and water sensors, with GPS tracking (Nazri et al., 2019). Other designs move the sensing off the cane. A backpack-based system covers the head-to-hip region and detected obstacles 94.4% of the time in a user test, but it did not do well with objects that did not face the sensor squarely (Kurz et al., 2019). Messaoudi et al. (2022) review many more tools for indoor and outdoor use and compare the different ways they find the user's location and give feedback. Khan et al. (2018) reviewed technology-assisted white canes and discussed what should be done next.

Even with all this work, a gap remains. Kim and Cho (2013) found that a smart cane avoided obstacles better than a white cane, but users also met usability problems, and smart canes cost around US$200, which most people in low-resource settings cannot pay. The prototype authors themselves list what is still missing. Sin et al. (2024) say their device must be lighter, respond faster and give different alerts for head-level and trunk-level hazards, and Nazri et al. (2019) say the detection range must be wider. The gap is therefore not a lack of technology. It is a lack of an affordable, light and reliable way to warn users in real time about the different hazards they meet outdoors, one that works in unstructured surroundings. Whether it is carried, worn or attached to something the user already owns is a choice for the concept stage.

1.2.2 Hypoglycaemia and glucose screening

Hypoglycaemia is dangerous because the brain depends on a constant glucose supply, and prolonged low glucose can cause neurological injury. In newborns it is described as the most common endocrine abnormality in children and is linked to increased illness and death (Mukunya et al., 2020). A community-based study of 1,416 newborns in Lira District, Northern Uganda, found that 2.2% had a blood glucose below 47 mg/dl; low values were associated with delayed initiation of breastfeeding and with an age of three days or less (Mukunya et al., 2020). The authors also point out that there is no consensus on the cut-off that separates harmless transient dips from pathological hypoglycaemia, and that cut-offs developed elsewhere may not suit African newborn-care practices.

Two lessons follow for design. First, risk is concentrated in identifiable groups and time windows, so screening that is rapid and repeatable at the bedside is valuable. Second, glucose measurement in routine practice commonly depends on enzymatic test strips, whose supply and cost can be barriers in low-resource facilities. This tension between clinical need and local supply is one that our concept generation stage will have to confront.

1.2.3 Simulation-based training in cardiac surgery

Cardiac surgery saves the lives of newborns with congenital heart defects and of young people with rheumatic heart disease, yet the workforce is thin. In 2017 about 12,000 adult and 4,000 paediatric cardiac surgeons practised worldwide, and the number of cardiac surgeons per million people was 140 to 180 times higher in high-income than in low-income countries (The Heart Surgery Forum, 2026). Training opportunities are limited and uneven, and the cost of training abroad deters many candidates from LMICs.

Regional examples show the scale of the problem. In Kenya, an estimated 80% to 90% of children who need cardiac surgery do not receive it, and only four to five heart surgeons are trained per year against a demand of about 20 (Business Daily Africa, 2022). Simulation is widely regarded as an effective way to build skills before operating on patients, but commercial simulators are expensive and many training institutions in lower-income countries cannot afford them; low-cost improvised models have been proposed as a workaround (Interactive CardioVascular and Thoracic Surgery, 2023). A low-cost, reproducible simulator for fundamental cardiac skills therefore addresses a genuine capacity gap.

1.3 Problem statement

Clinicians in resource-limited settings face problems that current technology does not solve, or solves at a cost and level of complexity that the health system cannot afford. A student design team also has little time, a small budget and only basic tools for building and testing. Choosing the wrong problem wastes a whole year. Choosing a well-supported one gives us a chance to build a prototype that matters.

So the problem in this report is about selection. Given a large pool of clinical needs collected in Ugandan healthcare settings, how can a team find the ones that are clinically important, poorly solved today and possible to engineer within one academic year? This report answers with a written, criteria-based screening process. It ends with these priority needs:
1.Visually impaired individuals navigating low-resource outdoor environments need real-time detection of environmental hazards to support independent mobility.
2.Patients at risk ofhypoglycaemia need rapid blood glucose screening to prevent neurological damage.
3.Cardiac surgical trainees need affordable simulation tools to practise fundamental cardiac surgery skills.
We analyse the first need in more detail below, in the order: need statement, condition fundamentals, existing solutions, stakeholders and market.

1.3.1 Need statement

Visually impaired individuals navigating low-resource outdoor environments need real-time detection of environmental hazards to support independent mobility.

The population is visually impaired individuals who travel outdoors in low-resource settings. The problem is that they need to learn about hazards in time, and not after they touch them. The

outcome is independent mobility, which here means travelling without a sighted guide and without injury. The statement does not say whether the answer should be worn, carried or attached to a cane, so the concept stage stays open.

1.3.2 Condition fundamentals

Visual impairment limits how a person finds hazards, judges distance and keeps a safe path. Messaoudi et al. (2022) describe how visually impaired people struggle to go outdoors and to protect themselves from moving and stationary objects, and how they lose confidence because their movement is restricted. The hazards fall into four groups.

Ground-level hazards. These include drop-offs, uneven ground and puddles. Hertel et al. (2019) added a laser sensor for ground-level obstacles, and Nazri et al. (2019) added water detection to their cane because puddles are a real outdoor hazard.

Body-level hazards. Sin et al. (2024) list tables as an example of trunk-level obstacles that the white cane does not detect. Kim and Cho (2013) note that cane users find it hard to detect protruding bars.

Head-level hazards. Tree branches and similar objects are missed by the cane (Sin et al., 2024). Manduchi and Kurniawan (2011) found that about 13% of blind and legally blind adults collide with head-level obstacles once a month or more, and that about 23% of such accidents lead to medical consequences.

Moving hazards. Vehicles cannot be detected until they are dangerously close, because the cane senses only about 1 m around the user (Singh et al., 2010; Kim & Cho, 2013).

The result is a mix of injuries from collisions and falls, a loss of confidence, and less independence. A useful solution therefore has to cover several hazard types and give the warning early enough for the user to react.

1.3.3 Existing solution analysis

The white cane is cheap, light and foldable, and it is the most widely used mobility aid in developing countries (Kim & Cho, 2013; Chanana, 2020). Its limits are clear. It detects obstacles only up to knee level and within about 1 m, so raised obstacles and moving vehicles are noticed too late (Singh et al., 2010).

Smart canes add electronics to the cane. The STIC cane combines an ultrasonic sensor for head-level obstacles with a laser sensor for ground-level ones and passes the distance to the user through vibration (Hertel et al., 2019). A retractable ultrasonic add-on detects head-level and trunk-level obstacles within 200 cm and 300 cm, but it needs to be lighter, respond faster and give different alerts for the two levels (Sin et al., 2024). The Smart e-Stick adds water detection and GPS tracking, and its authors want a wider detection range (Nazri et al., 2019). On the commercial side, the WeWALK Smart Cane 2 AI alerts the user to overhead obstacles through haptic and audible feedback, and it is priced at £849 before VAT in the RNIB shop (RNIB, n.d.). Kim and Cho (2013) found that smart canes avoided obstacles better than a white cane but also had usability problems, and that they cost around US$200.

Some solutions move the sensing off the cane. A backpack-based system covers the head-to-hip region and detected obstacles 94.4% of the time in a user test, but it did not do well with objects that did not face the sensor squarely (Kurz et al., 2019). Messaoudi et al. (2022) and Khan et al. (2018) review many more tools and show how wide the field is.

Taken together, the technology to sense hazards exists. What is missing is a version that is affordable, light, robust and tested in unstructured outdoor conditions. Chanana (2020) makes the same point: most devices are made for structured environments and often fail in unstructured ones.

1.3.4 Stakeholder analysis

This analysis comes from the literature and our own reasoning. We have not yet interviewed these groups, and Section 3.2 plans that validation.

Primary stakeholders are the visually impaired people who would use the device, since their safety and independence are at stake, and the family members, friends and sighted guides who may guide them today.

Secondary stakeholders are the people and places around the user: orientation and mobility trainers, teachers and staff at schools and rehabilitation centres for blind people, and eye-care workers who meet visually impaired people early.

Tertiary stakeholders shape the wider setting. They include government bodies responsible for health and disability, organisations of and for blind people, standards and regulatory bodies, funders such as WHO and UNICEF that work on assistive technology access, and local electronics suppliers and fabricators who would build and repair a device.

1.3.5 Market analysis

One industry report values the global market for assistive technologies for the visually impaired at USD 3,910 million in 2024 and forecasts USD 10,998 million by 2032, a growth rate of 13.8% a year (Credence Research, n.d.). Market reports differ in method, so we treat this only as a rough guide to the size of the field.

The picture is different in low-resource settings. WHO and UNICEF (2022) report that access to assistive products can be as low as 3% of the need in the poorest countries. A white cane is inexpensive (Kim & Cho, 2013), while electronic aids cost from around US$200 (Kim & Cho, 2013) to £849 before VAT (RNIB, n.d.), and Singh et al. (2010) describe internationally available devices as unaffordable for most users. A large segment is therefore left without a low-cost, robust device made for its conditions. Likely buyers are the users themselves, their families, schools and rehabilitation centres, non-governmental organisations and public programmes, and the right price point will have to be tested with them.

1.4 Objectives and scope of this report

Main objective: To identify and prioritise clinical needs suitable for a biomedical engineering design project, using the Stanford Biodesign needs-finding and needs-screening method.

Specific objectives:

- To select strategic focus areas and carry out background research on them.
- To gather clinical observations and interviews and convert them into well-formed need statements.
- To screen the need statements through four successive filters and justify each decision.
- To select the highest-priority needs and document the evidence behind them.

Scope: This report covers the Identify phase of Biodesign only. Concept generation, concept selection and prototyping were not reached this semester. They are described in Section 3.2 as the planned next stage.

CHAPTER TWO: NEEDS FINDING AND NEEDS SCREENING

2.1 The Stanford Biodesign process

The Biodesign process organises medical technology innovation into three phases (Yock et al., 2015). In the **Identify** phase, teams choose areas of focus, observe clinical practice, and write and screen need statements. In the **Invent** phase, they generate and select concepts and plan how a chosen concept would be developed. In the **Implement** phase, they develop the product, plan regulatory and reimbursement strategy, and prepare to bring the technology to patients. This semester's work sits entirely within Identify.

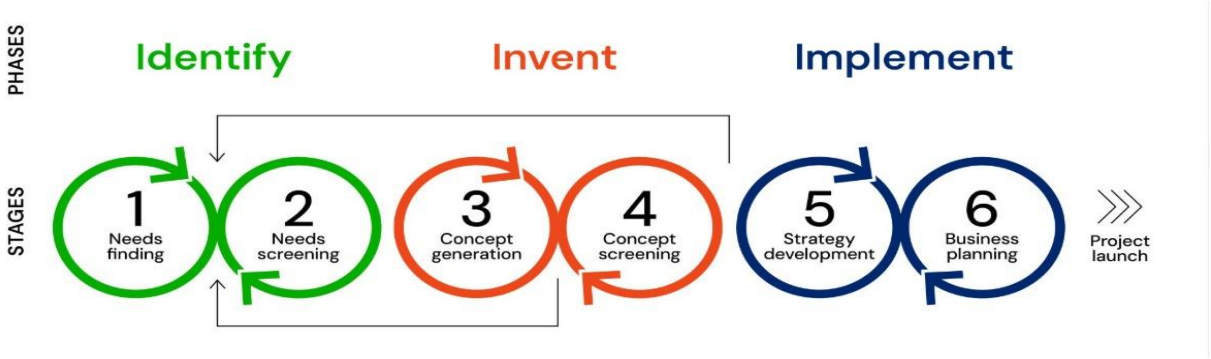


Figure 1 The Biodesign process

2.2 Needs finding

2.2.1 Choosing strategic focus areas

Open-ended searching for needs is inefficient. We therefore began by choosing focus areas where we could realistically gain clinical access and where unmet needs were likely, each paired with the clinicians whose work we wanted to understand (Table 1).

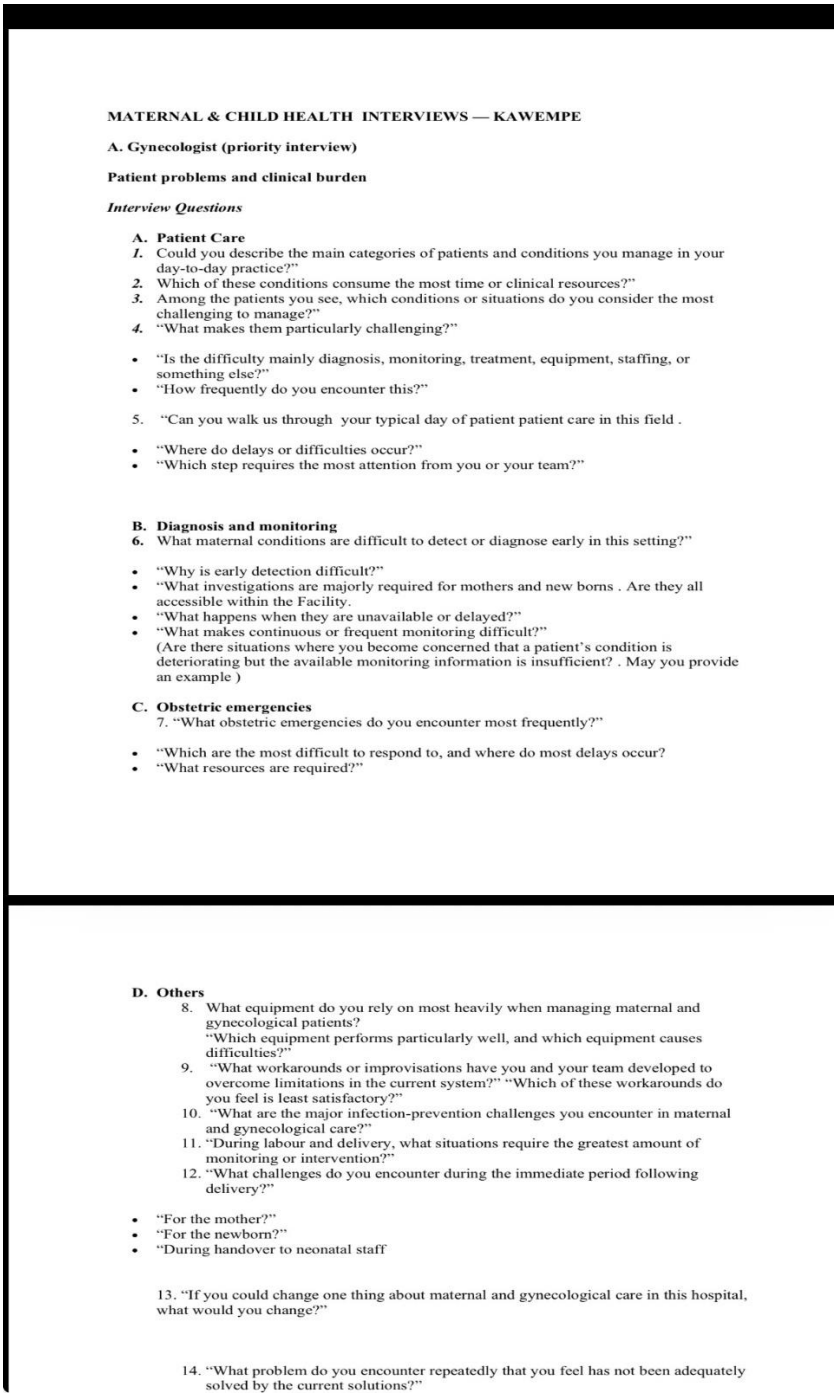
No.	Strategic focus area	Target clinicians
1	Maternal and neonatal child healthcare	NICU nurse, paediatrician, gynaecologist
2	Laboratory medicine	Laboratory technologist
3	Surgery	General surgeon
4	Rehabilitation	Biomedical engineer, orthopaedic surgeon

Table 1 Strategic focus areas and target clinicians

2.2.2 Background research and interview guides

Before going into the field, we reviewed background material on each focus area to understand the care environment, common workflows and known problems. This preparation meant that our questions in the field were informed rather than generic. We then drafted interview guides for each clinician group. The guides kept conversations focused while leaving room for follow-up questions, and they captured early hypotheses about possible needs to confirm or discard in practice.

MATERNAL & CHILD HEALTH INTERVIEWS — KAWEMPE



MATERNAL & CHILD HEALTH INTERVIEWS — KAWEMPE

A. Gynecologist (priority interview)

Patient problems and clinical burden

Interview Questions

A. Patient Care

1. Could you describe the main categories of patients and conditions you manage in your day-to-day practice?"
2. Which of these conditions consume the most time or clinical resources?"
3. Among the patients you see, which conditions or situations do you consider the most challenging to manage?"
4. "What makes them particularly challenging?"

- "Is the difficulty mainly diagnosis, monitoring, treatment, equipment, staffing, or something else?"
- "How frequently do you encounter this?"

5. "Can you walk us through your typical day of patient patient care in this field .

- "Where do delays or difficulties occur?"
- "Which step requires the most attention from you or your team?"

B. Diagnosis and monitoring

6. What maternal conditions are difficult to detect or diagnose early in this setting?"

- "Why is early detection difficult?"
- "What investigations are majorly required for mothers and new borns . Are they all accessible within the Facility.
- "What happens when they are unavailable or delayed?"
- "What makes continuous or frequent monitoring difficult?"
  (Are there situations where you become concerned that a patient's condition is deteriorating but the available monitoring information is insufficient? . May you provide an example)

C. Obstetric emergencies

7. "What obstetric emergencies do you encounter most frequently?"

- "Which are the most difficult to respond to, and where do most delays occur?
- "What resources are required?"

D. Others

8. What equipment do you rely on most heavily when managing maternal and gynecological patients?
   "Which equipment performs particularly well, and which equipment causes difficulties?"
9. "What workarounds or improvisations have you and your team developed to overcome limitations in the current system?" "Which of these workarounds do you feel is least satisfactory?"
10. "What are the major infection-prevention challenges you encounter in maternal and gynecological care?"
11. "During labour and delivery, what situations require the greatest amount of monitoring or intervention?"
12. "What challenges do you encounter during the immediate period following delivery?"

- "For the mother?"
- "For the newborn?"
- "During handover to neonatal staff

13. "If you could change one thing about maternal and gynecological care in this hospital, what would you change?"

14. "What problem do you encounter repeatedly that you feel has not been adequately solved by the current solutions?"

Figure 2 image of questionnaire used at Kawempe referral hospital

2.2.3 Observations and interviews

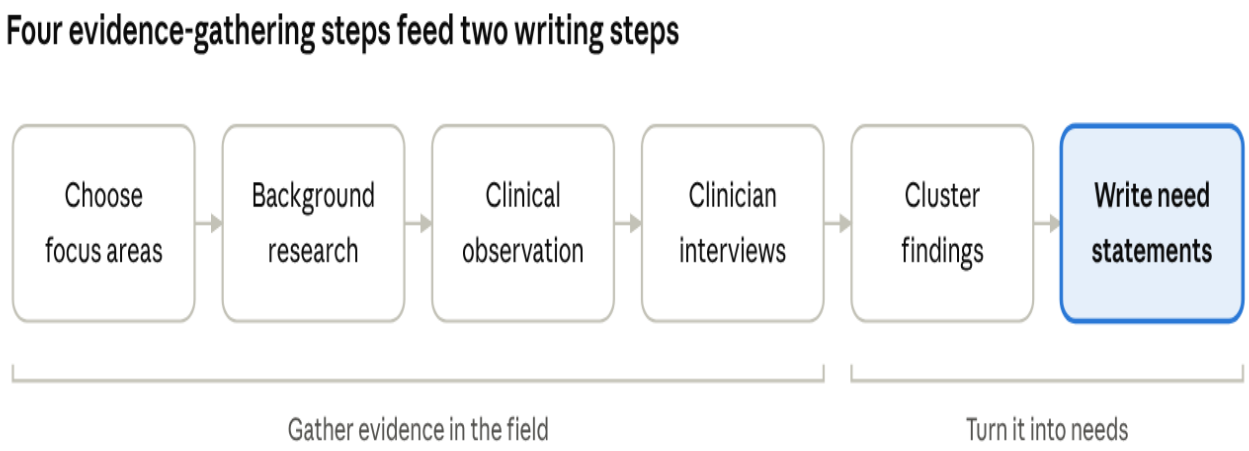
We combined interviews with observation of practice because clinicians often cannot describe a workaround they no longer notice, whereas an observer sees it immediately. Table 2 lists the clinicians who gave us interviews.



Figure 3 interview of the laboratory attendant at makerere

Clinician	Profession	Institution
Laymond Kalema	Laboratory technologist	Makerere University College of Health Sciences
Nakalanzi Sauds	Midwife	Kawempe National Referral Hospital
Eng. Allan Rodin Gidoi	Biomedical engineer	Ruby Hospital

Table 2 Clinicians interviewed during fieldwork



needs-finding workflow · 6 steps

Figure 2: Needs-finding workflow followed by Team Avron

2.2.4 Clustering findings and writing need statements

After fieldwork we organised our notes before writing any statement. Related observations and recurring complaints were grouped into **need clusters**, and the strongest clusters were then highlighted as **main needs**. This step stopped us from over-weighting a single vivid anecdote and helped us see which problems appeared in several settings.

Each finding was then written as a need statement with three components (Table 3). Writing in this form forces precision and keeps the statement free of any embedded solution, which preserves the freedom to invent later. Our process produced roughly 100 statements; the refined master list is reproduced in Appendix A.

Component	Meaning	Example (from our top three)
Population	The person or group experiencing the problem	Patients at risk of hypoglycaemia
Problem	What they need to do, achieve or manage	need rapid blood glucose screening
Primary outcome	The result that makes the need important	to prevent neurological damage

Table 3 Components of a need statement, with an example

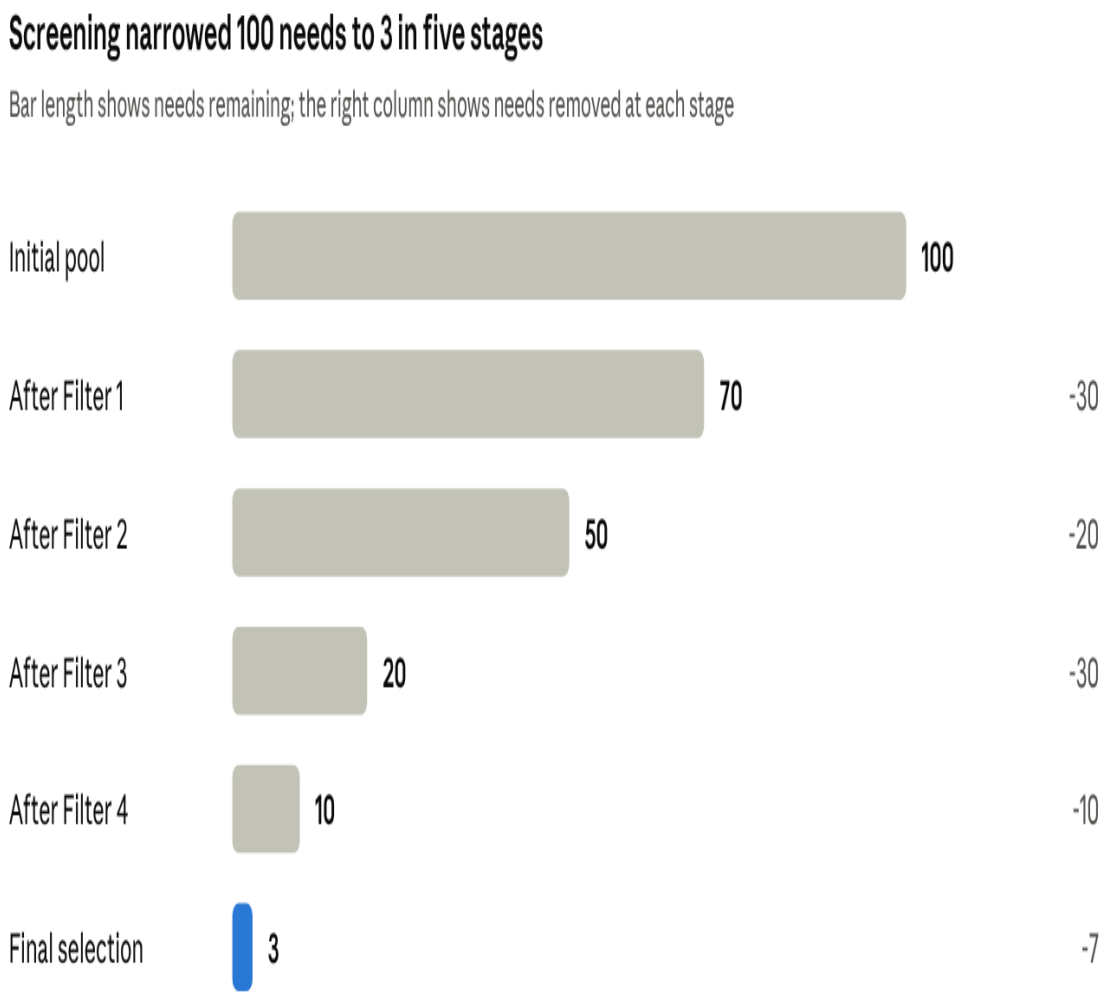
2.3 Needs screening

2.3.1 Overview of the screening funnel

A list of about 100 needs is far too many for one team to pursue. We reduced it through four filters, each with its own lens and its own rule for removing needs, so that every elimination could be explained by a stated criterion rather than by preference. Table 4 and Figure 3 summarise the funnel.

Stage	Screening lens	Needs in	Needs out	Needs retained
Filter 1	Regulatory, safety and clinical viability	100	30	70
Filter 2	Prototyping and technical feasibility	70	20	50
Filter 3	Stanford 2x2 importance versus satisfaction	50	30	20
Filter 4	Weighted decision matrix (MAUT)	20	10	10
Final selection	Team ranking and discussion	10	7	3

Table 4 Summary of the four screening stages



Team Avron screening records · 100 needs to final 3 (Table 4)

Figure 3: The screening funnel, from initial needs to final selection

2.3.2 Filter 1: regulatory, safety and clinical viability (100 to 70)

The first filter removed needs whose solutions would be unrealistic for a student team on regulatory or safety grounds. It applied a knockout rule: any need was discarded if it required a high-risk regulatory pathway comparable to FDA Class III implants, multi-year clinical trials, exposure to hazardous radiation, advanced genetic or molecular manipulation, or complex health-system policy change (Table 5). Needs that were really administrative or software-only problems, rather than device problems, were also removed at this stage.

Criterion	Question asked	If "yes"
Regulatory class	Is the solution likely to be a Class III device needing multi-year clinical trials?	Discard
Safety and ethics	Does testing require high voltage, radiation or invasive tissue contact?	Discard

Table 5 Filter 1 knockout rule and criteria

2.3.3 Filter 2: prototyping and technical feasibility (70 to 50)

The second filter asked whether the core of a solution could be built and tested with tools available to us: standard microcontrollers such as ESP32 and Arduino, off-the-shelf biomedical sensors, 3D printing and local mechanical fabrication. Four weighted criteria were scored from 1 to 5 (Table 6). A mandatory gate also applied. A need was removed regardless of its score if it depended on specialised microfluidic assays, imported biochemical reagents, complex optical or laser arrays, or long-term clinical trials for first validation.

Criterion	Weight	High score (4 to 5) means
Prototyping and fabrication feasibility	30%	Buildable with 3D printing, local fabrication, microcontrollers and standard sensors
Benchtop validation capability	30%	Testable on an electronics bench, mechanical rig or tissue phantom
Local resource and supply chain fit	20%	Uses locally available components and materials
Pre-clinical and clinical setup complexity	20%	Proof of concept needs only bench testing or short simulator validation

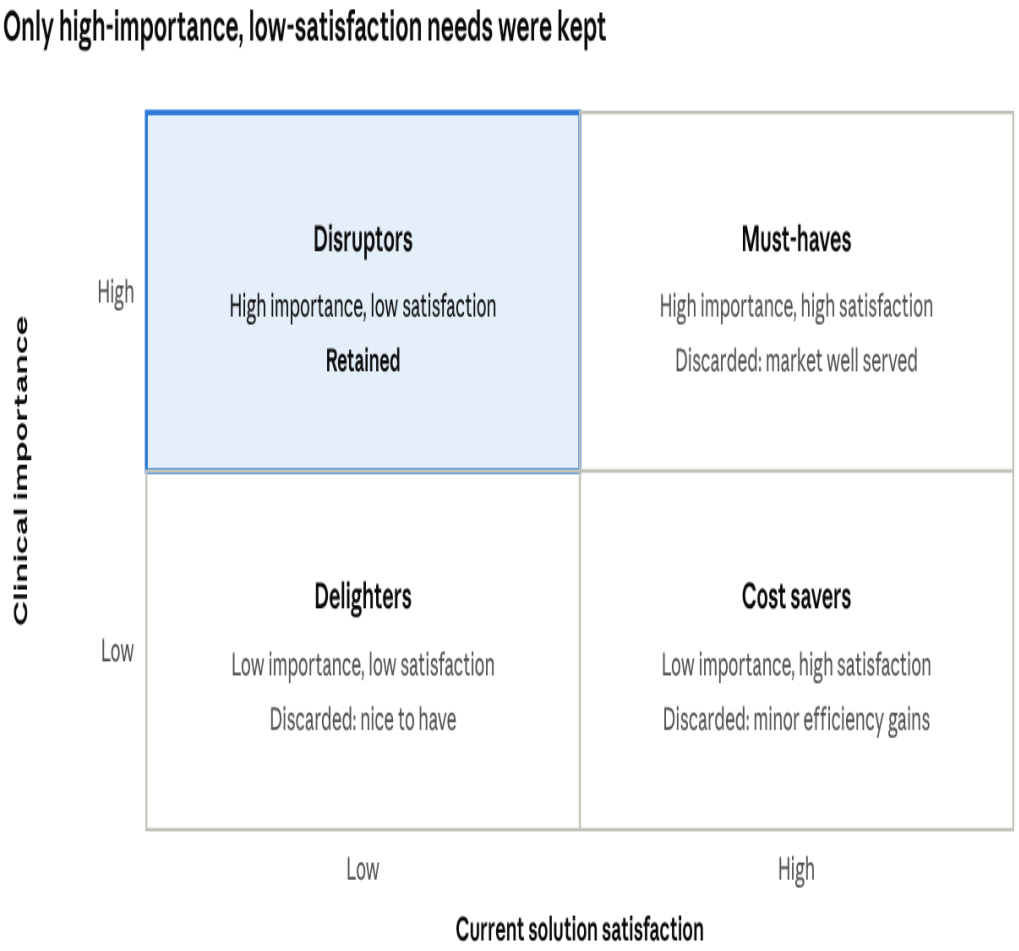
Table 6 Filter 2 criteria and weights

2.3.4 Filter 3: the Stanford 2x2 matrix (50 to 20)

The third filter used the Stanford 2x2 matrix, which plots each need by **clinical importance** (the severity and urgency of the outcome if the need is unmet) against **current solution satisfaction** (how well existing tools already solve it). The matrix sorts needs into four quadrants (Table 7). We kept only the **Disruptors**, where the need is serious and existing solutions are weak, because these offer the greatest room for a new device to make a difference. Our team plotted the candidates on the grid during a working session (Figure 4).

Quadrant	Importance	Satisfaction	Decision
Disruptors	High	Low	Retained
Must-haves	High	High	Discarded: market already well served
Cost savers	Low	High	Discarded: minor efficiency gains
Delighters	Low	Low	Discarded: nice to have, little life-saving value

Table 7 The four quadrants of the Stanford 2x2 matrix



Stanford 2x2 matrix · 4 quadrants, Disruptors retained

Figure 4: Stanford 2x2 matrix of clinical importance versus current solution satisfaction

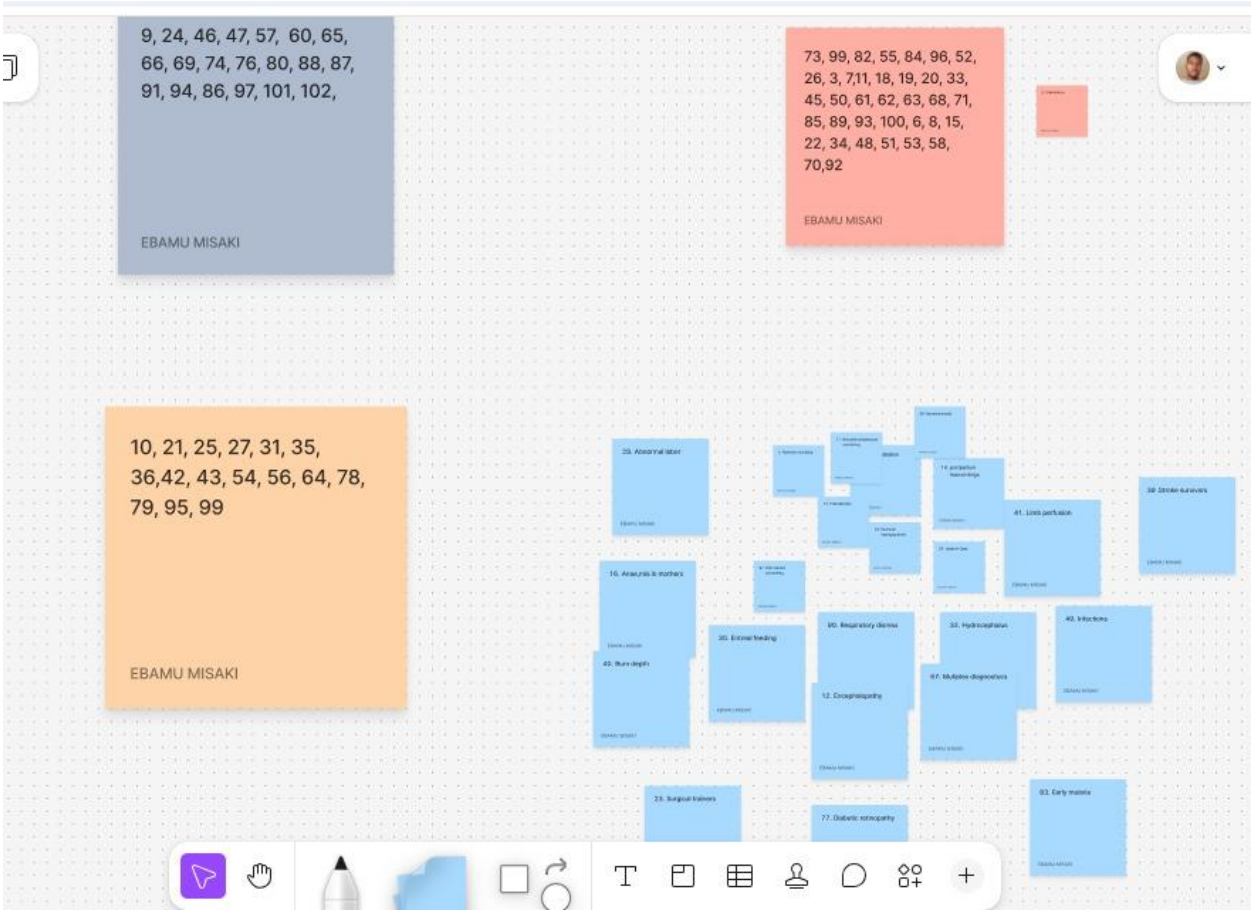


Figure 4 identification of disruptors, must haves, delighters and cost savers from our needs

2.3.5 Filter 4: weighted decision matrix (20 to 10)

The last filter was a multi-attribute utility (MAUT) matrix, a standard multi-criteria decision method (Belton & Stewart, 2002; Dieter & Schmidt, 2013). Each of the 20 remaining needs was scored from 1 (poor) to 5 (excellent) on six criteria, and the weighted total was calculated as:

Score = 0.20 (Market gap) + 0.25 (Urgency) + 0.15 (Prevalence) + 0.25 (Feasibility) + 0.05 (Safety) + 0.10 (LMIC relevance)

Urgency and feasibility carry the largest weights because a need must matter clinically and also be achievable by us within the year. Safety has a small weight because the major safety screening had already happened at Filter 1.

Criterion	Weight	What it measures
Market gap	0.20	How poorly existing products serve the need
Urgency	0.25	Severity of consequences if the need stays unmet
Prevalence	0.15	How many patients or users are affected
Feasibility	0.25	Whether a prototype can be completed in two semesters
Safety	0.05	Risk during development and testing
LMIC relevance	0.10	Fit with low-resource settings in Uganda

Table 8 Filter 4 criteria and weights

Table 9 applies these weights to the ten needs retained at this stage. The sub-scores are those recorded in the team's mid-semester presentation. The totals were recomputed from them with the formula above, and the needs are ranked by recomputed total.

Rank	Need	Mkt	Urg	Prv	Feas	Saf	LMIC	Total/5.00
6	Neonatal temperature monitoring	4.8	4.7	4.2	4.5	4.8	4.8	4.43
4	Cervical dilation measurement	4.8	4.8	4.8	4.0	4.2	4.5	4.54
3	Respiratory distress monitoring	4.8	4.7	4.5	4.0	4.5	4.5	4.45
5	Neonatal sepsis screening	4.8	4.9	4.5	3.8	4.5	4.5	4.48
1	Obstacle detection for the blind	4.5	4.7	4.5	4.2	4.5	4.2	4.84
2	Neonatal hypoglycaemia	4.5	4.8	4.2	3.5	4.2	4.2	4.65
7	Postpartum haemorrhage	4.8	4.8	4.5	3.0	4.5	4.2	4.230
9	Stroke survivors rehabilitation tool	4.2	3.8	4.0	4.5	4.8	4.0	4.155
10	Point-of-care cardiac screening	4.2	4.5	4.0	3.8	4.5	4.0	4.140
3	Surgical trainers	3.8	3.0	4.2	4.8	5.0	4.0	4.60

Table 9 MAUT scores of the ten shortlisted needs (total out of 5.0)

2.3.6 The final top ten needs

After reviewing the scored shortlist in Table 9, the team settled on the final ten needs listed in Table 10. They span neonatal care, diagnostics, surgical training, rehabilitation and assistive technology, which shows that the process did not collapse into a single clinical area.

No.	Need statement
1	Visually impaired individuals navigating low-resource outdoor environments need real-time detection of environmental hazards to support independent mobility.
2	Patients at risk of hypoglycaemia need rapid blood glucose screening to prevent neurological damage.
3	Infants with respiratory illness need rapid detection of acute respiratory distress to enable timely oxygen therapy.
4	Cardiac surgical trainees need affordable simulation tools to practise fundamental cardiac surgery skills.
5	Neonates in low-resource NICUs need rapid point-of-care screening for neonatal sepsis to enable timely antibiotic therapy.
6	Stroke survivors with upper-extremity paresis need targeted exercise tools to restore functional hand grasp.
7	Children undergoing standing therapy need postural stabilization during weight-bearing exercises to prevent fall injuries.
8	Neonates receiving phototherapy need secure ocular shielding to prevent retinal damage.
9	Patients experiencing significant haemorrhage need a rapid method to quantify blood loss to enable timely haemodynamic stabilisation.
10	Newborns at risk of congenital heart defects need point-of-care cardiac screening to enable early specialist referral.

Table 10 Team Avron's final top ten needs

2.3.7 Choosing the top three needs

From the ten, the team selected the three needs that offered the best combination of clinical significance, weak current solutions and feasibility for a one-year project. Table 11 summarises each and the evidence supporting it.

Priority	Need statement	Key evidence	Current gap	Engineering opportunity
1	Visually impaired individuals navigating low-resource outdoor environments need real-time detection of environmental hazards to support independent mobility.	About 43.3 million people were blind in 2020; about 13% of blind adults hit head-level obstacles monthly or more, and about 23% of such accidents cause medical consequences (GBD 2019 Collaborators, 2021; Manduchi & Kurniawan, 2011).	White canes detect only ground-level obstacles within about 1 m; smart canes cost around US$200 (Kim & Cho, 2013).	Low-cost sensor add-on that warns the user, for example by vibration, of head- and knee-level hazards beyond cane reach.
2	Patients at risk of hypoglycaemia need rapid blood glucose screening to prevent neurological damage.	2.2% of 1,416 newborns in Northern Uganda had glucose below 47 mg/dl; risk was linked to delayed breastfeeding and age of three days or less (Mukunya et al., 2020).	No agreed cut-off; screening is not always rapid or accessible; strip-based methods depend on consumable supply.	Rapid, low-cost bedside glucose screening suited to low-resource facilities.
3	Cardiac surgical trainees need affordable simulation tools to practise fundamental cardiac surgery skills.	Cardiac surgeons per million are 140 to 180 times more available in high-income than low-income countries (The Heart Surgery Forum, 2026); Kenya trains 4 to 5 heart surgeons a year against demand of about 20 (Business Daily Africa, 2022).	Commercial simulators are costly and out of reach for many training institutions (Interactive CardioVascular and Thoracic Surgery, 2023).	Affordable, reproducible physical simulator with measurable skill feedback.

Table 11 The three prioritised needs and their evidence base

CHAPTER THREE: DISCUSSION, CONCLUSION AND WAY FORWARD

3.1 Discussion

The screening funnel did what it was designed to do. Starting from about 100 statements, each stage applied a different lens, so that a need had to be safe, buildable, important and unsatisfied, and also well matched to our timeline, before it advanced. Because every knockout rule was written down in advance, the eliminations can be defended and revisited.

Several limitations deserve honest mention.

- **Fieldwork breadth.** Interviews covered three professionals across the focus areas. This is enough to seed needs but not enough to confirm them. Needs such as hazard detection for visually impaired people and cardiac surgical training also drew heavily on literature rather than direct observation, so they require clinical and user validation.
- **Scoring subjectivity.** Importance, satisfaction and utility scores come from team judgement. Sensitivity checks on the weights and independent re-scoring by several members would strengthen confidence in the ranking.
- **Broad need statements.** "Patients at risk of hypoglycaemia" covers newborns, diabetic patients and others, whose clinical contexts differ. Narrowing the population will be essential before concepts are generated.
- **Supply constraints.** Conventional glucose measurement relies on enzymatic test strips, which conflict with our local-supply criterion. The concept stage must weigh this directly.

The selected needs also illustrate a useful spread of device types: assistive sensing, point-of-care diagnostics and training hardware. Each is plausible with microcontrollers, standard sensors and local fabrication.

3.2 Way forward: concept generation

The next stage follows the Invent phase of Biodesign.

1. **Clinical and user validation.** Interview clinicians, visually impaired users, trainers and surgical trainees to confirm and refine each need and to establish what a successful solution must achieve.
2. **Design criteria.** Convert the validated needs into measurable, weighted design criteria.
3. **Concept generation.** Run structured brainstorming and build a morphological matrix of sub-functions and options.
4. **Concept screening and selection.** Score concepts against the criteria, using a Pugh matrix and further stakeholder input, and select one project for design synthesis.
5. **Benchtop prototype.** Build and test a proof-of-concept prototype using phantoms and bench rigs.

3.3 Conclusion

This report shows how Team Avron moved from fieldwork to a justified shortlist. Four strategic focus areas guided our background research, observations and interviews, which produced a pool of need statements. Four screening filters then reduced that pool to ten needs and finally to three: real-time hazard detection for visually impaired people, rapid glucose screening for patients at risk of hypoglycaemia, and affordable cardiac surgical simulation. Each is supported by evidence of clinical importance and by gaps in existing solutions. Concept generation was not reached this semester. Completing it, beginning with validation of the three needs, is the next milestone toward a single, feasible design project.

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APPENDIX A: MASTER LIST OF REFINED NEED STATEMENTS

The refined master list produced from our needs-finding work (version of 18 September 2026) is reproduced below in the team's own wording. The list contains 103 statements.

1. Visually impaired individuals navigating low-resource outdoor environments need real-time detection of environmental hazards to support independent mobility.
2. Healthcare workers in low-resource facilities need a rapid method to decontaminate reusable medical equipment to reduce cross-contamination.
3. Patients undergoing surgery need an affordable infection barrier method to reduce surgical-site infection rates.
4. Medical imaging technologists need personal radiation shielding to reduce cumulative occupational radiation exposure.
5. NICU healthcare workers need an efficient method to sterilize reusable medical tubing to reduce nosocomial infections.
6. Newborns with gastroschisis need rapid protection of exposed abdominal organs to prevent fatal infection.
7. High-risk surgical and critical care patients need an affordable method to maintain normothermia to prevent perioperative complications.
8. Radiographers in low-resource facilities need affordable digital imaging capture tools to lower radiography costs.
9. Children with mobility impairments need growth-accommodating assistive devices to maintain functional independence.
10. Children undergoing standing therapy need postural stabilization during weight-bearing exercises to prevent fall injuries.
11. Surgical patients need an affordable method to maintain body temperature to prevent perioperative hypothermia.
12. Neonates with hypoxic-ischemic encephalopathy need affordable therapeutic hypothermia to reduce brain injury.
13. Pregnant women at risk of preeclampsia need affordable continuous blood-pressure monitoring to detect hypertension early.
14. Patients experiencing significant hemorrhage need a rapid method to quantify blood loss to enable timely hemodynamic stabilization.

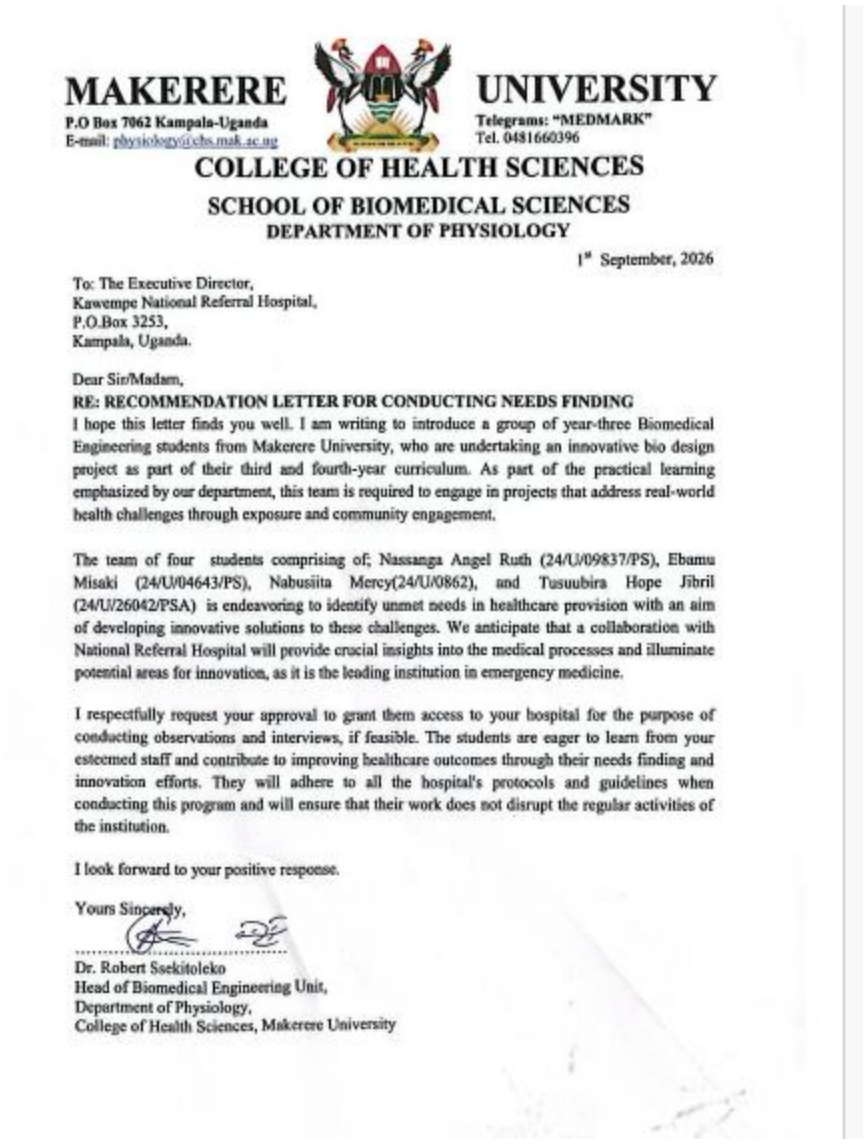
15. Patients at high risk of gastric cancer need non-invasive screening for early mucosal dysplasia to improve survival rates.
16. Pregnant women in underserved areas need low-cost screening to detect maternal anemia early.
17. Patients at risk of hypoglycemia need rapid blood glucose screening to prevent neurological damage.
18. Neonates receiving phototherapy need secure ocular shielding to prevent retinal damage.
19. Patients requiring urinary catheterization need secure catheter stabilization to prevent accidental displacement.
20. Patients with acute burns need extended-wear wound dressings to minimize dressing-change pain.
21. Elderly individuals with gait instability need continuous fall-prevention assist tools to prevent fall injuries.
22. Patients in cardiac arrest need continuous high-quality chest compressions to improve resuscitation survival.
23. Surgical trainees need affordable practical simulation tools to accelerate procedural skill acquisition.
24. Laboratory technicians performing genetic testing need rapid sequencing protocols to reduce turnaround time.
25. Manual wheelchair users traversing slopes need a method to prevent unintended rolling to reduce tipping and ejection accidents.
26. Patients with diabetic foot ulcers need early detection of tissue inflammation to prevent deep infection.
27. Newborns in regional hospitals need objective point-of-care hearing screening to identify congenital hearing loss early.
28. Cardiac surgical trainees need affordable simulation tools to practice fundamental cardiac surgery skills.
29. Women experiencing prolonged labour need objective tracking of labour progression to detect dystocia early.
30. Neonates receiving enteral tube feeding need non-invasive verification of tube positioning to prevent pulmonary aspiration.
31. Children immobilized in casts need pressure monitoring beneath cast linings to prevent early tissue breakdown.
32. Children with hydrocephalus and shunts need non-invasive detection of shunt failure to enable prompt surgical referral.
33. Neonates receiving NG feeding need early detection of tube displacement to prevent pulmonary aspiration.
34. Patients recovering from major surgery need early detection of surgical wound dehiscence to enable immediate repair.

35. Patients undergoing prolonged surgery need continuous pressure distribution monitoring over vulnerable anatomy to prevent pressure injuries.
36. Patients with reduced lower-limb sensation need pressure monitoring beneath orthotic devices to prevent pressure ulcers.
37. Newborns at risk of congenital heart defects need point-of-care cardiac screening to enable early specialist referral.
38. Neonates in low-resource NICUs need rapid point-of-care screening for neonatal sepsis to enable timely antibiotic therapy.
39. Stroke survivors with upper-extremity paresis need targeted exercise tools to restore functional hand grasp.
40. Patients with acute burns need accurate assessment of burn depth to guide surgical debridement.
41. Trauma patients with suspected limb ischemia need rapid limb perfusion assessment to prevent tissue necrosis.
42. Cancer patients receiving chemotherapy need continuous nutritional tracking to detect malnutrition early.
43. Women undergoing IUD insertion need instant verification of correct device placement to prevent uterine perforation.
44. Newborns at risk of hypoglycemia need rapid blood glucose screening to prevent neurological damage.
45. Pregnant women living with HIV need continuous PMTCT tracking to prevent vertical HIV transmission.
46. Pediatric patients with physical disabilities need growth-accommodating mobility devices to maintain functional independence.
47. Children with clubfoot undergoing casting need growth-accommodating serial casts to maintain anatomical correction.
48. Trauma patients at risk of compartment syndrome need non-invasive compartment pressure monitoring to enable early fasciotomy.
49. Patients with suspected infection in low-resource settings need rapid differential testing between bacterial and viral pathogens to guide antibiotic stewardship.
50. Patients with suspected multi-pathogen infections need low-cost multiplex testing from a single sample to accelerate diagnosis.
51. Patients requiring urgent blood transfusion need rapid point-of-care blood compatibility verification to prevent hemolytic transfusion reactions.
52. Patients with suspected tuberculosis in rural areas need rapid point-of-care TB confirmation to initiate early therapy.
53. Cancer patients receiving chemotherapy need early infection detection to enable prompt treatment of neutropenic fever.
54. Cancer patients with chemotherapy-induced peripheral neuropathy need objective symptom monitoring to detect worsening nerve damage.

55. High-risk pregnant women in rural areas need continuous remote monitoring between antenatal visits to detect complications early.
56. Women living with obstetric fistula need accessible screening tools to enable early surgical referral.
57. Blood-bank staff need automated tracking of blood-product shelf life to minimize blood wastage.
58. Critically ill ICU patients need rapid point-of-care blood-gas analysis to guide immediate clinical decisions.
59. Premature and low-birth-weight infants need continuous thermal monitoring to detect hypothermia early.
60. Patients with chronic wounds in low-resource settings need low-cost wound management materials to accelerate healing.
61. Newborns in low-resource settings need affordable point-of-care sickle-cell screening to enable early diagnosis.
62. Elderly patients with hypertension need affordable home blood-pressure devices to facilitate regular monitoring.
63. Individuals with limb loss need affordable prosthetic devices to restore functional mobility.
64. Cancer patients in underserved settings need accessible psychosocial support tools to improve treatment adherence.
65. Patients receiving repeated chemotherapy need affordable long-term central venous access to reduce peripheral venipunctures.
66. Newborns in low-resource hospitals need low-cost precision weighing to guide nutritional management.
67. Patients in limited-lab settings need low-cost multiplex tests to detect multiple infections from one sample.
68. Healthcare workers in low-resource settings need affordable reusable PPE to maintain infection-control protection.
69. Children with hearing loss in low-resource settings need affordable, durable hearing aids to improve communication.
70. Individuals with upper-limb amputations need an affordable prosthetic control system to restore multi-articulating hand movement.
71. Neonates in transit need non-electric thermal maintenance to prevent hypothermia.
72. Newborns in hospitals without pediatric cardiologists need accessible CHD screening to enable early referral.
73. Pregnant women attending early antenatal care need accessible gestational age estimation to identify high-risk pregnancies early.
74. Children under five in drought-affected areas need rapid screening for severe acute malnutrition to enable timely nutritional rehabilitation.
75. Communities in cholera-prone areas need rapid testing for drinking water contamination to identify unsafe water sources.

76. Patients with diabetic foot ulcers need affordable offloading solutions to relieve wound pressure and promote healing.
77. People with diabetes need accessible retinal screening to detect diabetic retinopathy early.
78. Elderly patients in routine care need brief cognitive screening tools to detect early cognitive decline.
79. People affected by leprosy with reduced foot sensation need early detection of pressure injuries to prevent foot ulceration.
80. Stroke survivors living at home need accessible home rehabilitation tools to maintain long-term recovery.
81. People recently exposed to HIV need high-sensitivity early detection screening to enable prompt antiretroviral therapy.
82. Neonates with severe hyperbilirubinemia need non-invasive bilirubin measurement to reduce painful blood draws.
83. Patients with early malaria infection need low-density parasite detection to enable early treatment.
84. Patients undergoing major surgery need real-time blood loss measurement to guide fluid management.
85. Patients requiring venipuncture need affordable vein visualization to increase first-attempt success rates.
86. Biomedical engineers in low-resource hospitals need supply-chain access to local equipment replacement parts to reduce equipment downtime.
87. Laboratory staff in unstable grid regions need diagnostic equipment with power backup to prevent testing interruptions.
88. Clinical engineering teams need predictive maintenance monitoring for medical devices to eliminate unexpected equipment failure.
89. Patients receiving IV infusions need safe fluid warming prior to administration to maintain core temperature.
90. Infants with respiratory illness need rapid detection of acute respiratory distress to enable timely oxygen therapy.
91. Sterile processing staff need a scale-prevention method for steam sterilizers to maintain sterilizer uptime.
92. Surgical teams need automated surgical instrument tracking to prevent retained surgical items.
93. Women with prolonged second-stage labour need assisted fetal descent to prevent labor complications.
94. Laboratory technicians preparing microscopy slides need standardized slide preparation tools to improve diagnostic consistency.
95. Newborns exposed to meconium need early detection of meconium aspiration syndrome to enable immediate airway intervention.

96. Patients undergoing major surgery need early detection of blood-volume loss to prevent hypovolemic shock.
97. Patients receiving spinal anesthesia need precise anesthetic volume delivery monitoring to eliminate dosing errors.
98. Infants with congenital heart defects need portable cardiac monitoring to detect early deterioration.
99. Neonates receiving IV therapy need early detection of fluid extravasation to prevent tissue necrosis.
100. Patients arriving at emergency departments need rapid triage categorization to prioritize critically ill patients.
101. Premature infants post-discharge need remote physiological monitoring to enable early intervention for clinical decline.
102. Patients undergoing rehabilitation need objective functional progress measurement to guide physical therapy decisions.
103. Neonatal intensive care units need an affordable method to continuously monitor patient blood oxygen saturation to prevent hypoxia-induced organ damage.



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COLLEGE OF HEALTH SCIENCES

SCHOOL OF BIOMEDICAL SCIENCES

DEPARTMENT OF PHYSIOLOGY

1st September, 2026

To: The Executive Director,
Kawempe National Referral Hospital,
P.O.Box 3253,
Kampala, Uganda.

Dear Sir/Madam,

RE: RECOMMENDATION LETTER FOR CONDUCTING NEEDS FINDING

I hope this letter finds you well. I am writing to introduce a group of year-three Biomedical Engineering students from Makerere University, who are undertaking an innovative bio design project as part of their third and fourth-year curriculum. As part of the practical learning emphasized by our department, this team is required to engage in projects that address real-world health challenges through exposure and community engagement.

The team of four students comprising of; Nassanga Angel Ruth (24/U/09837/PS), Ebamu Misaki (24/U/04643/PS), Nabusiita Mercy(24/U/0862), and Tusuubira Hope Jibril (24/U/26042/PSA) is endeavoring to identify unmet needs in healthcare provision with an aim of developing innovative solutions to these challenges. We anticipate that a collaboration with National Referral Hospital will provide crucial insights into the medical processes and illuminate potential areas for innovation, as it is the leading institution in emergency medicine.

I respectfully request your approval to grant them access to your hospital for the purpose of conducting observations and interviews, if feasible. The students are eager to learn from your esteemed staff and contribute to improving healthcare outcomes through their needs finding and innovation efforts. They will adhere to all the hospital's protocols and guidelines when conducting this program and will ensure that their work does not disrupt the regular activities of the institution.

I look forward to your positive response.

Yours Sincerely,

.....

Dr. Robert Ssekitoleko
Head of Biomedical Engineering Unit,
Department of Physiology,
College of Health Sciences, Makerere University

Figure 5 sample recomendation letter for hospital visit